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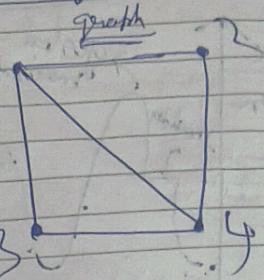
Constraint Satisfaction Problem

Domain = {R, G, B}

V = {1, 2, 3, 4}

no adjacent vertice of same color

C = {1≠2, 1≠3, 3≠4, 2≠4, 1≠4}



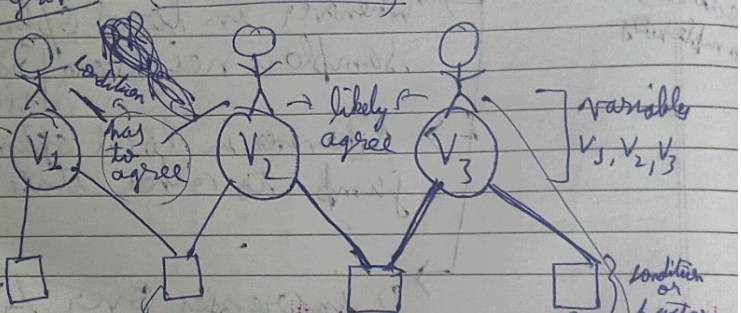
constraint — straight cut way

	1	2	3	4
Initial domain	RGB	RGB	RGB	RGB
1 = R	R	GB	GB	GB
2 = G	R	G	GB	B
3 = B	R	G	B	
3 = G	R	G	G	B

Not possible backtrack

different problem — assume only B, R

CSP representation — factor graph



x_1	$f_1(x_1)$
R	0
B	1

$f(x_1) = [x_1 = B]$

x_1	x_2	$f_2(x_1, x_2)$
R	R	1
R	B	0
B	R	0
B	B	1

x_2	x_3	$f_3(x_2, x_3)$
R	R	2
R	B	1
B	R	1
B	B	2

fixed

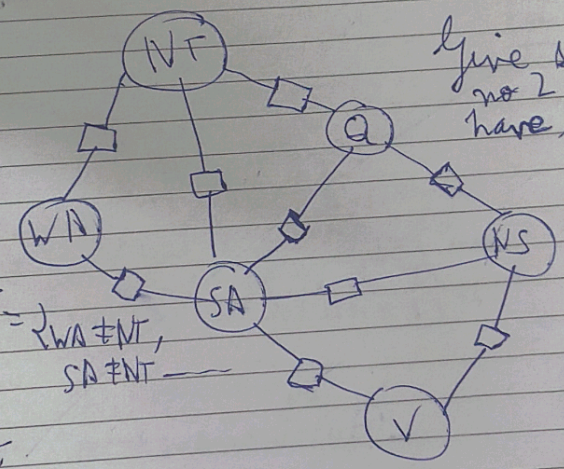
x_3	$f_3(x_3)$
R	2
B	1

we have defined in pairwise factor graph

x_1	x_2	x_3	weight	
R	R	R	0, 1, 2, 2	= 0
R	R	B		= 0
R	B	R		= 0
R	B	B		= 0
B	R	R	1, 0, 2, 2	= 0
B	R	B	1, 0, 1, 1	= 0
B	B	R	1, 1, 1, 2	= 2
B	B	B	1, 1, 2, 1	= 2

multiply their weight, if it is > 0

Q- Eg



Give such that no 2 adj. vertices have same color.

Constraint = $WA \neq NT$, $SA \neq NT$ ——

List

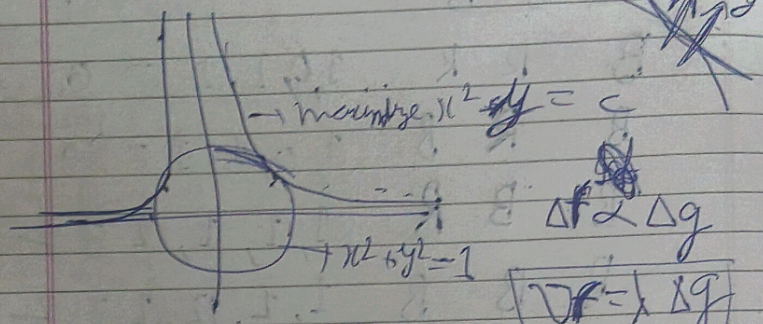
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~~Constraint optimization~~

Constraint optimization problem

Eg- Maximize $f(x,y) := x^2 \cdot y$

On the set $\underbrace{x^2 + y^2 = 1}_{g(x,y)}$.



$\Delta f \parallel \Delta g$

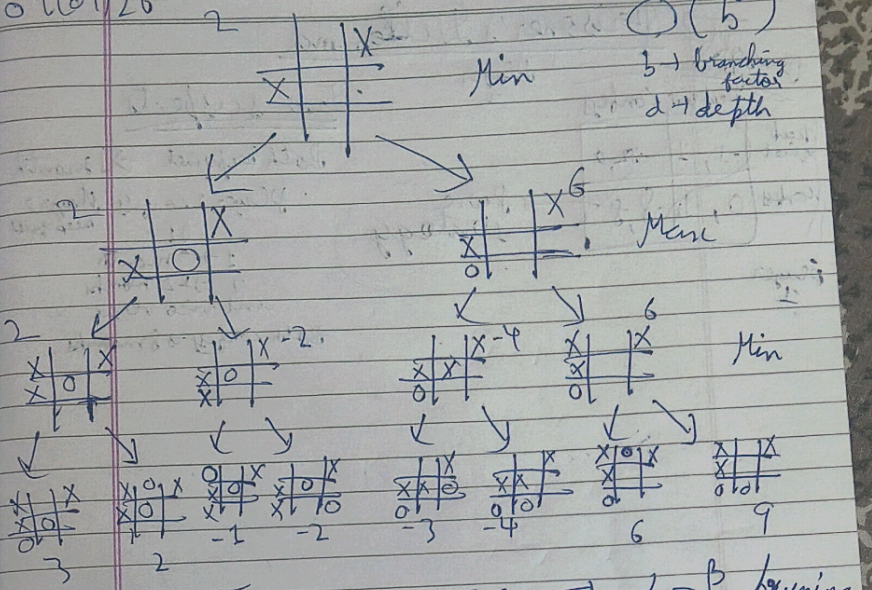
$\boxed{\nabla f = \lambda \Delta g}$

Min-Max algorithm

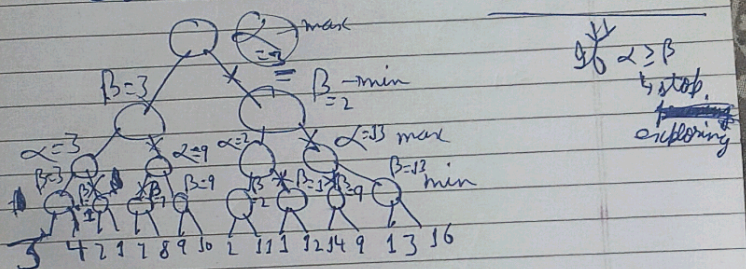
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$O(b^d)$
$b \rightarrow$ branching factor
$d \rightarrow$ depth



To reduce T.C. → α-β pruning



If $\alpha \geq \beta$ stop pruning

$\alpha \rightarrow$ maximizes
$\beta \rightarrow$ minimizes

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Prisoner's Dilemma

Two suspects

Payoff matrix

Player 1 \ Player 2	keep quiet	confess
keep quiet	-1, -1	-12, 0
confess	0, -12	-8, -8

→ Pure strategy

- Both keep quiet → 1 month
- Player 1 confess, Player 2 keep q
  - 1 ⇒ 0 months
  - 2 ⇒ 12 months
  - and vice versa
- Both confess ⇒ 8 months